

Phase 7: Project Documentation

ComicCraft - AI Comic Story Creator using Gemini Models

Comprehensive Project Technical Report

Abstract

ComicCraft is an advanced web-based application that leverages generative artificial intelligence to democratize comic book creation. Built on the FastAPI framework and integrated with Google's Gemini LLMs (Gemini 1.5 Flash and Gemini 1.5 Pro) alongside Diffusion-based visual inking models, ComicCraft transforms simple natural language premises into complete, sequentially structured 5-panel comic strips. The system automatically performs narrative structuring, character dialogue synthesis, ambient scene captioning, concurrent illustration generation, and publication-ready multi-page PDF compilation. The result is an accessible, intuitive platform enabling non-artists, educators, and storytellers to produce print-quality graphic narratives in seconds.

1. Introduction

Storytelling through sequential comic panels combines literary pacing with visual immersion. However, the high barrier to entry—demanding both creative writing prowess and skilled illustration—restricts visual storytelling primarily to professional artists. ComicCraft overcomes this barrier by creating an automated end-to-end multimodal pipeline where text models and visual generators cooperate synergistically under the orchestration of a lightweight Python backend.

2. System Architecture & Methodology

The architecture is structured into four primary layers:

- Presentation Layer:** Built with semantic HTML5, modern CSS featuring halftone comic styling, and Jinja2 templating for server-rendered dynamic UI.
- Controller & API Layer:** Powered by FastAPI, exposing form-handling routes (/generate), headless JSON endpoints (/generate-comic/json), and direct file download streams.
- Generative AI Core:**
 - Gemini 1.5 Flash:** Outlines a 5-panel narrative arc (Threshold, Rising Action, Confrontation, Climax, Resolution) formatted strictly as a JSON array.
 - Gemini 1.5 Pro:** Expands the outline into an authentic comic book script, formatting ambient environmental audio into CAPTION:, narrator voice into NARRATION:, and distinct character dialogue lines.
 - Diffusion Inking Engine:** Employs prompt-sanitized concurrent multi-threading to ink 768x512 panel illustrations in parallel.
- Publishing Layer:** The layout_builder.py binds textual elements and imagery, and exporters.py compiles a 5-page A4 PDF using FPDF2 with Unicode TrueType typography.

3. Implementation Modules

Module Name	File Location	Key Responsibility
gemini_flash.py	app/gemini_flash.py	generate_outline(user_prompt)
gemini_pro.py	app/gemini_pro.py	generate_story(outline)
image_generator.py	app/image_generator.py	generate_image(prompt, filename)
layout_builder.py	app/layout_builder.py	build_comic_layout(images, story, outline)
exporters.py	app/exporters.py	save_pdf(layout)
routes.py	app/routes.py	Route handling & ThreadPool coordination
main.py	app/main.py	Server entry point & static directory mounting

4. Results & Deliverables

- Interactive Creation Interface:** Form with quick-fill presets (The Brave Fox, Cyberpunk Megacity, Cosmic Odyssey, Magic

Academy).

2. Sequential Story Preview: Dynamic comic preview with yellow caption boxes, speech bubbles, and illustration frames.
3. Print-Ready PDF Publication: Multi-page PDF output timestamped and organized with exactly 1 panel per page.
4. Download & Export Confirmation: Seamless user feedback flow redirecting to /export-success.

5. Conclusion & Future Scope

ComicCraft demonstrates that multimodal AI orchestration can democratize visual narrative arts. Future iterations will explore multi-character consistency training (via LoRA), multi-page story arcs, voice dubbing/audiobook generation, and interactive digital comic reader modes.